

PAPER_882: Expansion Lagrangian Euler-Lagrange

Author: Daniel T. Murphy – Star Magic / UQFF Framework **Date:** 2026-04-08 **Session:** 209 **Source:** Sessions 204-208 standalone module integration **Calculator:** Expansion-LagrangianEulerLagrangeCalc (CP4 #466) **CVW:** v2.0.0 compliant

Abstract

Variational principle applied to the positive buoyancy expansion scalar. $L_{\text{expansion}} = E^+(t) \cdot V \cdot S_{26}([SSq])$ with Euler-Lagrange variation $\delta S / \delta \varphi_{\text{expansion}} = 0$ providing closure for the expansion potential. Links the expansion Lagrangian to the buoyancy ratio and filament volume.

1. Core Equations

$$L_{\text{expansion}} = E^+(t) \cdot V \cdot S_{26}$$
$$\delta S / \delta \varphi = \partial L / \partial \varphi - d/dt(\partial L / \partial \dot{\varphi}) = 0$$

2. Parameters

Parameter	Default	Description
E_0	1.0 J	Initial energy scale
t	0.0 s	Time parameter
V_filament	1e48 m ³	Filament volume
F_UBi_over_FU	1.1	Buoyancy-to-field ratio

3. UQFF Integration

This calculator integrates `positive_et_expansion.py` from Session 205 into the CP4 pipeline as class #466.

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